

Surgical EQ

Six precise parametric bands on a large curve with a live spectrum, mid / side per band and dynamic bells.



Overview

Surgical EQ is a clean, precise equaliser for detailed work: finding and taming resonances, carving space, and shaping the mid and side of a stereo signal separately. Six bands sit on a large curve display over a live spectrum of the output, and you can work directly on the curve by dragging the band dots.

Each band can be a bell, a shelf, a notch or a steep 24 dB per octave cut, and can work on the stereo pair, the mid only or the side only. Bells can also be dynamic: they only move when the band gets loud, so a harsh resonance is cut only when it rings. A high-pass and low-pass filter with selectable slopes sit ahead of the bands.

Reach for it for

- Notching out ringing resonances, hum or a single harsh frequency

- Taming harshness or boom only when it happens (dynamic bells)
- Mid / side work: cleaning the low end of the sides, lifting the centre
- Steep clean-up filtering with the HPF / LPF at up to 48 dB per octave
- Finding problem frequencies with SOLO and the analyser

Controls

The display

The curve covers 20 Hz to 20 kHz and plus or minus 24 dB. Each active band is drawn faintly in its own colour, with the total curve on top. When a band is dynamic, a second total curve in the accent colour shows where the EQ is right now. Behind it, the filled spectrum shows the output after the EQ.

Band dots	Each active band has a numbered dot in its colour (round for bells and notches, square for shelves and cuts). Drag a dot to move the band: left and right for frequency, up and down for gain. Hold Shift for fine moves. The dot shows M or S under it when the band works on the mid or side.
Mouse wheel on a dot	Changes the band's Q.
Double-click a dot	Switches that band off.
Double-click empty space	Switches on the first band that is off, as a bell at that frequency and gain.

Band

This section shows the selected band. Clicking or dragging a dot also selects its band.

1 - 6	The six coloured keys pick the band to edit.
Type	Click to choose: OFF, BELL, LOW SHELF, HIGH SHELF, NOTCH, LOW CUT or HIGH CUT. The cuts are 24 dB per octave with no bump at the corner. All six bands start as bells at 0 dB, at 80 Hz, 250 Hz, 800 Hz, 2.5 kHz, 6 kHz and 12 kHz.

SOLO	Hear only the region the selected band works on: a band-pass at its frequency and Q, on its target (stereo, mid or side). Use it to hunt for a resonance, then switch it off. While a band is soloed, the FLAT ALL key in the header becomes SOLO OFF.
FREQ	The band's frequency, 20 Hz to 20 kHz.
GAIN	Boost or cut, -24 to +24 dB. Bells and shelves only - notches and cuts have no gain.
Q	The band's width, 0.1 to 30. Low values are broad and gentle; high values are very narrow, for single resonances. On shelves it sets the shape of the corner (0.3 at least). The cuts have a fixed 24 dB per octave slope instead.
ST / M / S (TARGET)	What the band works on: ST the stereo pair, M the mid (what is common to both sides), S the side (the difference).

Dynamics

For bells only (the section shows BELLS ONLY for other types). A dynamic bell listens to its own band: when the level there goes over THRESH, the band moves away from its GAIN setting by up to RANGE. The further over the threshold, the further it moves - half of RANGE at 6 dB over.

DYN	Makes the selected bell dynamic.
THRESH	The level in the band where it starts to move, -60 to 0 dB.
RANGE	How far the band can move, -24 to +24 dB. Negative cuts a resonance only when it rings; positive lifts the band only when it gets busy.
ATTACK	How fast dynamic bands react, 0.5 - 100 ms. Shared by all bands.
RELEASE	How fast they recover, 10 - 1000 ms. Shared by all bands.

Filters

The high-pass and low-pass come before the six bands. Both are Butterworth filters, with no bump at the corner.

HPF	High-pass filter, 20 Hz (OFF - out of circuit) to 1 kHz.
LPF	Low-pass filter, 1 kHz to 20 kHz (OFF at the top).
SLOPE	Click to step through x1, x2, x3 and x4: 12, 24, 36 or 48 dB per octave. Both start at x2 (24 dB). The key lights while its filter is in.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Click the TRIM knob (without moving it) to switch the guard; its light sits at the knob's upper right and is lit while it is on. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. The light glows brighter while it is catching peaks.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping.

Master

OUTPUT	Output level, -24 to +12 dB.
OUT meters	Output level, left and right.

The header

Theme picker	Changes the panel's colour theme.
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S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Next to the S / M / L size keys. Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW	Switches between the hardware-style panel and the flat look.
ANALYSER	Shows or hides the spectrum behind the curve.
FLAT ALL	Sets the gain of all six bands to 0 dB (types, frequencies and Q are kept). While a band is soloed this key reads SOLO OFF and ends the solo instead.

Tips

- To find a resonance: boost a narrow bell (Q 8 or more) by 10 dB or so, sweep it until the problem jumps out, then turn the gain down into a cut.
- For a resonance that only rings on some notes, use a dynamic bell with GAIN at 0 and a negative RANGE, and set THRESH so it only moves on the loud notes.
- Cleaning up a stereo mix: a LOW CUT on S at 100 - 150 Hz keeps the low end centred without touching the bass in the middle.
- SOLO with a narrow Q is a quick way to hear exactly what you are about to cut.
- Double-click a dot to switch a band off and compare the sound with and without it.

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